Roop Yaswanth Nagabhairava

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RESEARCH EXPERIENCE

University of Florida, Smart Autonomy Lab Gainesville, FL
Research Engineer Aug 2026 - Present

- **Binary code similarity:** Co-developing a method to quantify whether a patched or recompiled binary preserves the original's functional and security behavior without source code, across compiler, optimization, and architecture variants.
- **Cross-embodiment manipulation:** Developing methods to transfer Vision-Language-Action (VLA) policies learned on one collaborative robot arm and gripper to different arms and gripper geometries; testing whether trajectory resampling can adapt existing demonstrations to a new robot, so they need not be recollected.
- **Hierarchical VLM-to-VLA control:** Designing a framework where a vision-language model turns complex instructions into simple prompts for a VLA policy, with a failure verifier, so a robot can finish multi-step tasks and recover.

PROJECTS

CineMatch | Hybrid Recommender with Graph Contrastive Learning roop-yaswanth.github.io/work/cinematch
Python, PyTorch, RecBole, BGE-M3, FAISS, SciPy

- Fused multilingual BGE-M3 semantic retrieval (1.37M titles) with XSimGCL graph-contrastive collaborative filtering (MovieLens-32M) and shipped it as a live Next.js/FastAPI product with rating-driven discovery and language controls.
- In a temporal evaluation ranking all 65,723 titles rather than a sampled list (2,500 users, 19,419 future positives), reached 0.066 NDCG@10, about twice MostPopular and LightGCN (Holm-corrected $p < 1e\text{-}29$).
- Semantic retrieval is the only path that reaches titles unseen in training, where collaborative models score exactly zero, and covers 3.4x more of the catalog at rank 10, so new releases can surface before anyone rates them.
- Language bridging and DPP raised cross-language exposure from 0.13 to 0.30 for 1,013 viewers with NDCG@10 unchanged, more than doubling the share outside their main language, at a 0.008 NDCG@10 cost across 2,500 users.

SoundVector | Contrastive Embeddings for Music Retrieval roop-yaswanth.github.io/work/soundvector
Python, PyTorch, hnswlib, scikit-learn, Gemini API

- Trained 256-d contrastive track embeddings for 899K songs and built HNSW retrieval with a ranker for artist variety and recording deduplication; on 1,000 random seeds it raised distinct artists per top 10 from 2.8 to 9.0 and cut lists repeating a recording from 35% to 0, turning queues dominated by the seed's artist into a rotation, at a cost in genre share (98% to 89%) and mean popularity (0.22 to 0.43).
- Built feedback-driven personalization in which active saves and completed previews shape the queue, removing a like rebuilds the profile, and dislikes filter recordings across catalog and provider IDs, so listeners can correct their profile.

Multi-Image Super-Resolution for Prostate MRI roop-yaswanth.github.io/work/mri
Python, PyTorch, MONAI, torchmetrics, pydicom, NumPy, scikit-image

- Built a multi-image super-resolution pipeline on 782 patients (TCIA Prostate-MRI-US-Biopsy) to restore the detail lost between thick MRI slices, without the longer scans that thinner slices require.
- Four-slice SwinIR reached 28.18 dB PSNR and 0.925 SSIM on 8,587 held-out slices, 1.6 dB above the next-best model and about half as variable; error maps showed ResNet blurring capsule edges and SRGAN adding texture not in the scan.

EDUCATION

University of Florida, Gainesville, FL May 2026
Master of Science in Applied Data Science | GPA: 3.90/4.00
Coursework: Applied Machine Learning I & II, Deep Learning in Medical Image Analysis, Math Foundations for Data Science I & II

K L University, Vijayawada, India May 2024
Bachelor of Technology in Artificial Intelligence and Data Science | GPA: 9.28/10.00

SKILLS

Research Areas: binary code similarity, program analysis, robot manipulation, multimodal vision-language and vision-language-action models, graph and contrastive representation learning, recommender systems
Machine Learning: PyTorch, Hugging Face Transformers, RecBole, scikit-learn, TensorFlow/Keras, MONAI, LoRA/QLoRA fine-tuning, FAISS, HNSW
Engineering: Python, SQL, TypeScript, Linux, FastAPI, PostgreSQL, MongoDB, NumPy, Pandas, SciPy
Cloud & Tools: AWS, Azure, Google Cloud, Docker, Git, GitHub Actions, Claude Code (agentic coding)
Certifications: Google Cloud Associate Cloud Engineer; Microsoft Azure AI Fundamentals (AI-900)